AUDIT · PRIOR ART · RESEARCH DESIGN · BLUEPRINT

AirGuard India 2.0

What the existing system does, what the rest of the field has already built, what is genuinely missing, and the experiments designed to test it.

47 external systems reviewed

5 pilot experiments

24 deliverables

Nothing claimed without a source

STATUS	**Evidence and design. No part of version 2.0 has been built.**
REVIEWED	07 October 2026
COURSE	AD23532 — Principles of Data Science
INSTITUTION	Rajalakshmi Engineering College (Autonomous, affiliated to Anna University, Chennai)
CURRENT SYSTEM	https://webapp-phi-taupe.vercel.app
REPOSITORY	https://github.com/nishaids/airguard-india

Contents

1. Summary
   What was done, six findings, the gap that survives, and where each deliverable is

2. Part 1 — Evidence
   Audit of the existing system · pilot experiments · 47-system prior-art corpus · gap matrix · hypothesis

3. Part 2 — Research design
   Data sources as verified · benchmark · evaluation · the risk-controlled advisory · advisory index

4. Part 3 — Platform architecture
   Batch-first architecture · API · database schema · MLOps · security

5. Part 4 — Product and interface
   Interface specification · wireframes · design system · roadmap to a service · demonstration

6. Part 5 — Programme
   Roadmap · paper outline · risks · limitations · acceptance criteria · decisions needed

Summary

What this document is

The brief asked for the existing AirGuard India system to be audited, compared with what the rest of the world has built, and redesigned into a research-grade platform, in that order, and without claiming novelty that the evidence does not support.

This document is the result of the first three steps of that order — **audit, prior art, research gap** — together with the full design for everything that follows. No part of AirGuard 2.0 has been built. The deployed application is unchanged.

Three kinds of statement appear here and are kept apart: what was **measured** (on the repository, the live site, or in a pilot experiment), what was **read from a cited source**, and what is **proposed**.

What was done

- The repository and the live deployment were audited, including latency, error handling and the correctness of what the site tells users.
- **Forty-seven external systems were reviewed**: government forecasting systems, commercial products, health-impact tools, methods papers, Indian forecasting papers and open student projects. Each is recorded with its source in `prior_art_corpus.csv`.
- **Every proposed live data source was tested with a real request.** Three work today without a key; one official source could not be reached.
- **Five pilot experiments** were run on the existing 2015–2020 record to check whether each gap named in the brief is real.

Six findings that change the plan

1. **None of AirGuard's features is new on its own.** Next-day forecasting, cohort-specific advice, explanation, exposure windows and percentage-reduction scenarios all exist in operational systems. India already has 400-metre physical-model forecasting for Delhi and district-level forecasts nationwide.
2. **The closest prior art is stronger than AirGuard's advisory index.** Forecaster.health issues forecast-driven mortality-risk alerts by age and sex across 31 European countries from epidemiological models. AirGuard's cohort weights are hand-set.
3. **The live site's "confidence" is wrong.** 41% of forecasts are shown as at least 95% confident; 89% of those are correct. The model that was swapped in to fit the hosting limit is the cause.

4. **Prediction intervals work on average and fail when the air is worst.** At 90% nominal coverage, three standard methods cover 57–75% of *Severe* days.
5. **Letting the model abstain when uncertain is unsafe.** It declines four in five *Severe* days, because uncertainty and pollution rise together. This idea from the brief is rejected on the evidence.
6. **Cross-city generalisation is a smaller problem than assumed.** Removing a city from training costs 3.5% in error on average.

The gap that survives

Of the 47 systems reviewed, eight combine a forecast with cohort advice and three combine a forecast with uncertainty. **None combines all three, and none attaches a stated error rate to the advice it gives.**

The pilot shows why that matters. An advisory issued from the point forecast fails to warn on 18% of polluted days. Issued from the 75th percentile it misses 9%, and over-warns on 24% of days instead of 14%. Where to sit on that curve is a decision about whose errors cost more, and it differs for a healthy adult and a cardiac patient.

The proposed research contribution is to make that decision explicit and controlled: each cohort's advisory is issued at a risk level calibrated to hold under-warning below a declared target on future data. Whether that is new to the literature is not yet established and is not claimed.

What happens next

Five decisions are needed before work continues (Part 5): API keys, a database, approval to correct the live site, a target date, and a reviewer for the health text. One of them is urgent: the high-pollution season begins within weeks, and data collection that starts after it has passed will not contain the severe days the study depends on.

Where each requested deliverable is

DELIVERABLE	LOCATION
A — Existing project audit	Part 1, A
B — Global prior-art analysis	Part 1, B; `prior_art_corpus.csv`
C — Competitive matrix	Part 1, B.2 and C
D — Research-gap matrix	Part 1, D
E — Research hypothesis	Part 1, E
F — Novelty statement	Part 1, F

DELIVERABLE	LOCATION
G — Architecture	Part 3, G
H — Dataset strategy	Part 2, H
I — Preprocessing pipeline	Part 2, I
J — Model benchmark plan	Part 2, J
K — Evaluation plan	Part 2, K
L — Advisory index methodology	Part 2, L
M — Interface redesign specification	Part 4, M
N — Page wireframes	Part 4, N
O — Design system	Part 4, O
P — Backend and API	Part 3, P
Q — Database schema	Part 3, Q
R — MLOps	Part 3, R
S — Implementation roadmap	Part 5, S
T — Research-paper outline	Part 5, T
U — Startup and SaaS roadmap	Part 4, U
V — Competition demonstration	Part 4, V
W — Risks and mitigation	Part 5, W
X — Acceptance criteria	Part 5, X

Part 1 — Evidence

What exists today, what others have already built, what is genuinely missing, and what the data in hand says about each proposed gap. Nothing in this part is a plan; every statement is either read from the repository, measured on the deployed system, measured in a pilot experiment, or taken from a cited source.

Status vocabulary used throughout the blueprint:

TAG	MEANING
Implemented	Exists in the repository and the deployment, and has been verified
Partial	Exists, but falls short of what its description implies
Proposed	Designed in this blueprint; not built
Future research	Requires data or validation that is not yet available

A. Existing system audit

A.1 What the system is

ITEM	FACT	SOURCE
Dataset	*Air Quality Data in India (2015–2020)*, CPCB data mirrored on Kaggle, licence CC0, SHA-256 recorded	`data/raw/PROVENANCE.json`
Temporal coverage	2015-01-01 to 2020-07-01. **Nothing after 1 July 2020 exists in the system.**	data card
Spatial unit	One value per city per day (mean over that city's stations). No station identity survives.	data card
Cities	26 in the source; **12 retained** by two pre-declared gates (≥ 80% AQI completeness, ≥ 730 labelled days)	data card
Rows	29,531 raw → 18,395 after cleaning	cleaning report
Pollutants	PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, benzene, toluene, xylene	config
Meteorology	**None.** No wind, temperature, humidity, rain or boundary-layer height	data card

ITEM	FACT	SOURCE
Features	178 candidates → 100 (regression) and 102 (classification) after selection: lags, rolling statistics, ratios, calendar terms, sub-indices, a city target encoding	feature report
Split	Chronological. Train 10,331 rows to 2019-05-25; test 4,397 rows from 2019-05-26 to 2020-06-30	evaluation files
Regression	Tuned LightGBM, MAE 15.05, RMSE 23.41, R^2 0.9125. Persistence baseline MAE 18.90 (−20.1%)	tuning report
Classification	Reported best: XGBoost + SMOTE, macro-F1 0.7411 (tuned 0.7452). **Deployed on Vercel: LightGBM + SMOTE, macro-F1 0.7145**	evaluation, deployment report
Clustering	K-Means, k = 2 on 12 cities, silhouette 0.355, reproduced by Ward linkage	clustering report
Explainability	TreeSHAP, global and one local example, precomputed	explainability report
Deployment	Next.js static pages plus two Python serverless functions on Vercel; trees exported to JSON and evaluated without ML libraries	`DEPLOY_VERCEL.md`

A.2 Capability audit

CAPABILITY	HOW IT WORKS TODAY	KEEP?	WEAKNESS FOUND	RECOMMENDED CHANGE
Provenance and reproducibility — Implemented	Checksummed acquisition, 20-stage pipeline, every reported number read from JSON	**Keep unchanged**	None material	Extend the same discipline to live data: every measurement carries source, station, timestamps and a quality flag
Event-aware cleaning, causal imputation — Implemented	Outlier rule exempts festival, stubble and lockdown windows; forward-fill chosen over a better-scoring non-causal method	**Keep**	Forward-fill is applied at city-day level only	Re-apply at station-hour level when station data arrives

CAPABILITY	HOW IT WORKS TODAY	KEEP?	WEAKNESS FOUND	RECOMMENDED CHANGE
Next-day regression — Implemented	LightGBM on today's pollutants and their history	**Keep as the baseline to beat**	MAE on *Severe* days is 58.8 with a bias of −22.6: the model under-predicts exactly the days that matter. No weather inputs. No interval	Add forecast meteorology; report by band; add calibrated intervals
Next-day band classification — Partial	Deployed model differs from the reported one	Replace	The deployed classifier is overconfident (see pilot P1). A separate classifier can contradict the regression	Derive the band and its probabilities from one calibrated predictive distribution
Persistence baseline and leakage demonstration — Implemented	Baseline scored on the same split; same-day leakage shown by experiment	**Keep**	—	Add CAMS as a second baseline: the forecast a user can already get for free
Health Advisory Index — Partial	`min(100, D + 0.25·C)` over cohort-weighted sub-indices; seven cohorts	Keep the idea, rebuild the index	Weights are hand-set, a fact the code states and the interface does not. **The index saturates:** on Delhi, 5 Nov 2019, all six vulnerable cohorts read exactly 100.0, so it cannot tell them apart on the days it exists for	Remove the ceiling; anchor to published guidance and Indian concentration–response evidence; publish a sensitivity analysis (Part 2, L)
Lowest-exposure window — Partial	Mean hourly profile per city and season from the historical hourly file	Keep the correction about hourly AQI	A climatology, not a forecast: it gives the same answer every day of a season. Labelled as a "cleanest window" without stating that air may still be unhealthy	Forecast tomorrow's hourly profile; always state the absolute level alongside the relative one
Policy simulator — Implemented	Re-scores the full record under a proportional concentration change	**Keep** — already labelled as a scenario, not a prediction	No uncertainty; no weather adjustment; days are counted, people are not	Add sensitivity bands, side-by-side scenarios, population weighting where census data supports it

CAPABILITY	HOW IT WORKS TODAY	KEEP?	WEAKNESS FOUND	RECOMMENDED CHANGE
Lockdown and festival analysis — Partial	Before/after comparison against a matched baseline	Keep as description	Not adjusted for weather, so it is not a causal estimate. The agreement between the simulator (−37.0%) and the lockdown (−34.0%) is a plausibility check, not a validation	Apply meteorological normalisation (Grange et al. 2018) before any causal wording
City archetypes — Partial	K-Means on pollutant-mix shape	Keep as exploratory	Twelve points; one cluster has three members. Too little data to support a general claim	Recompute at station level when a few hundred stations are available
SHAP explanations — Partial	Precomputed global ranking and one fixed local example	Keep method	Not computed for the forecast the user is looking at. Explanations are technical, not plain language	Per-forecast attribution, grouped into a few named drivers, with a plain-language sentence
Drift check — Partial	Flags inputs more than 3 SD from the training mean	Keep as one signal	On polluted days it fires constantly, because pollution *is* the tail (pilot P4)	Separate data-integrity failures from high-pollution conditions
Web application — Implemented	Five pages, responsive grid, dark theme	Rework information architecture	Replays historical dates without saying so prominently; English only (the Streamlit app had Tamil and Hindi); severity is conveyed mainly by colour	Part 4
API — Partial	Two POST endpoints	Extend	No versioning, rate limiting or schema validation; a malformed body returns HTTP 500 with the Python error text	Part 3

A.3 Corrections required before anything is submitted or presented

These are places where the project currently says more than its evidence supports. They are listed first because a reviewer who finds one will discount everything else.

1. **The live site's "confidence" is not a probability.** It shows the deployed classifier's top-class score, which is overconfident: 41.3% of test forecasts are displayed at 95% or higher, and 88.9% of those are correct (pilot P1).
2. **"Modelled with an explicit lockdown flag" is false.** Six event flags were offered to feature selection (`festival_flag`, `days_from_diwali`, `stubble_season_flag`, `lockdown_flag`, `post_lockdown_flag`, `covid_year_flag`); none was retained by either model. The sentence appears in the application and in the generated insights.
3. **The SHAP finding about the HAI is over-read.** The healthy-adult HAI ranks first, but it is an arithmetic re-expression of today's sub-indices, strongly collinear with AQI and PM2.5. Its rank shows how attribution is shared among correlated inputs. It is not independent evidence that the index is valid, and earlier project text that says so should be withdrawn.
4. **The site replays history.** Asked for today's date, the API silently substitutes 2020-07-01 and the page headlines "forecast for 2 July 2020". It must be labelled a historical replay until live data exists.
5. **The simulator's agreement with the lockdown is not a validation** (see the table above).
6. **The Tamil and Hindi advisory text is unreviewed.** It was machine-drafted. It is health advice, and it should not be shown to the public until a fluent reviewer has checked it.
7. **Cross-validation and test error disagree.** Four-fold expanding-window CV gives MAE 29.9 ± 10.0; the held-out test gives 15.1. The test year is cleaner than the training years and includes the lockdown, so the single-holdout figure flatters the model. Rolling-origin evaluation across several years is required.

A.4 Deployed system, measured on 7 October 2026

CHECK	RESULT
Forecast endpoint latency	4.9 s cold, 0.9 s warm
Policy endpoint latency	4.8 s (re-scores 18,395 rows on each call)
Unknown city	Handled: HTTP 400 with the list of covered cities
Malformed JSON body	**HTTP 500**, Python exception text returned to the caller
What-if input range	**Not validated.** A reduction of −500% is accepted and multiplies PM2.5 six-fold to 1,142 µg/m³, above the pipeline's own plausibility ceiling, and returns "99% confidence"
Security headers	HSTS present. No Content-Security-Policy, X-Frame-Options or X-Content-Type-Options. CORS is `*`
Languages	English only

CHECK	RESULT
Data freshness indicator	None

A+. Pilot evidence on the existing record

Before committing a roadmap to uncertainty, cross-city generalisation and abstention, each was tested on data already in hand (`python -m src.research.pilot_trust`, results in `results/pilot_trust.json`). These are pilots: historical data only, models refitted on a proper-training slice so that a calibration slice (18 Nov 2018 – 25 May 2019, 2,073 rows) stays unseen. They indicate where the real problems are. They are not results to publish.

P1. Is the displayed confidence honest?

CLASSIFIER	ACCURACY	MEAN CONFIDENCE	EXPECTED CALIBRATION ERROR	SHOWN AS ≥ 95%	CORRECT WHEN SHOWN ≥ 95%
LightGBM + SMOTE — what the live site serves	76.9%	86.2%	**0.093**	41.3% of forecasts	**88.9%**
XGBoost + SMOTE, tuned — the reported model	79.3%	79.7%	0.019	11.4% of forecasts	98.0%

The reported model is well calibrated. The one substituted for it to fit the serverless size limit is not. The substitution cost 2.7 points of macro-F1, which was disclosed, and also the meaning of the confidence figure, which was not known until now.

P2. Do distribution-free intervals reach nominal coverage?

Nominal coverage 90%, scored on the 4,397-row future test set.

METHOD	OVERALL COVERAGE	MEAN WIDTH (AQI POINTS)	*GOOD* DAYS	*SEVERE* DAYS (N = 44)
Split conformal, constant width	94.7%	96.1	99.3%	**56.8%**
Conformalised quantile regression	90.2%	67.5	79.0%	**75.0%**
Mondrian, by predicted band	91.0%	70.3	94.3%	**65.9%**

Marginal coverage is easy and is met. Coverage on the days that matter is not: no method reaches 90% on *Severe* days. An interval that is right on average and wrong when the air is worst is the specific failure a public advisory cannot afford. Forty-four *Severe* days is a small sample, which is itself a constraint the full study must address with more winters of data.

P3. How much is lost on a city the model has never seen?

For each city, a model was fitted on the other eleven and scored on the held-out city's test period, with the city-identity feature removed from both arms.

	MAE, CITY IN TRAINING	MAE, CITY UNSEEN	PERSISTENCE
Weighted mean over 12 cities	15.14	15.67 (**+3.5%**)	18.91
Worst case: Lucknow	20.76	24.08 (+16.0%)	28.13
Only failure: Thiruvananthapuram	8.77	9.16	8.79

The unseen-city model still beats persistence in 11 of 12 cities. **This gap is smaller than the mission brief assumed.** A pooled model built on each city's own recent history transfers well, because what it learns is short-term dynamics, not the city. Two cautions follow. First, an "unseen" city here still supplies its own lag history at prediction time; a city with no monitor at all is a different and much harder problem. Second, all twelve cities passed the same completeness gates. The stronger test is on the fourteen cities those gates excluded.

P4. If the model declines to answer when least certain, which days does it decline?

ABSTENTION RULE	DECLINED	MAE ON ANSWERED	SHARE OF *SEVERE* DAYS DECLINED	SHARE OF ALL POLLUTED DAYS DECLINED
None	0%	15.20	—	—
Widest 10% of intervals	10%	12.89	**81.8%**	54.1%
Widest 20% of intervals	20%	11.18	**100%**	91.8%
Highest 10% of relative width	10%	14.80	2.3%	2.5%
Most novel 10% of inputs (z-score)	10%	14.33	**97.7%**	36.7%

Abstaining on uncertainty improves average error and removes the forecast on four in five *Severe* days. Uncertainty and pollution are confounded: the model is least sure when the air is worst. **Naive abstention makes the system silent exactly when it is needed**, so it is rejected as a design. Abstention

is kept only for data-integrity failures (stale or missing sensors), where no forecast is the honest output.

P5. Does advising from a higher quantile fix under-warning?

The alternative to silence is caution: issue the advisory from an upper quantile of the predictive distribution instead of its centre. "Polluted" means the observed band was *Poor* or worse (523 test days).

ADVISORY ISSUED FROM	POLLUTED DAYS UNDER-WARNED	*SEVERE* DAYS UNDER-WARNED (OF 44)	DAYS OVER-WARNED	EXACT BAND
Point forecast	94 (**18.0%**)	8	13.7%	78.9%
65th percentile	78 (14.9%)	6	17.0%	77.3%
75th percentile	47 (**9.0%**)	5	23.7%	72.5%
85th percentile	24 (4.6%)	3	33.9%	64.4%
Upper bound of the 90% interval	10 (1.9%)	2	51.9%	47.6%

This is a clean, monotone trade-off between two kinds of error, and the point forecast sits at one arbitrary position on it: it misses nearly one polluted day in five. Choosing where to sit is a decision about whose mistakes are more costly, and that differs by cohort. This is the strongest signal in the pilot and it shapes the hypothesis in section E.

B. Prior-art analysis

B.1 Method and scope

Forty-seven external systems were reviewed on 7 October 2026: operational government systems, commercial products, health-impact tools, methods papers, Indian forecasting papers and open student projects. Each was located by web search and its facts taken from the cited page; the full record, with URLs, is `docs/airguard2/prior_art_corpus.csv`. This is a comparison corpus, **not a census of everything ever built**, and every "not found" below is scoped to these 47 sources.

<!-- generated by src/research/prior_art_tables.py from prior_art.json; do not edit -->

B.2 Capability matrix

● documented in the source reviewed · ◐ partial or generic · a dot means *not documented in the source reviewed*, which is not a claim of absence. Reviewed 2026-10-07.

SYSTEM	TYPE	COVERAGE	HORIZON	LIVE	FORE-CAST	WEA-THER	SUB-CITY	UNCER-TAINTY	COHORT ADVICE	ADVICE FROM FORECAST	EXPOSURE WINDOW	EXPLA-NATION	SCEN-ARIO	TRUST / ABSTAIN	UNSEEN CITY	OPEN
Google Maps Platform Air Quality API (current)	Commercial API	100+ countries	Hourly, up to 96 h	●	●	·	●	·	●	·	·	·	·	·	·	·
Google Air View+ (India) (2024)	Commercial product	India, 150+ cities	Real-time	●	·	●	●	·	·	·	·	·	·	·	·	·
SAFAR-India (IITM / IMD) (2010s onward)	Government operational	India: Delhi, Pune, Mumbai, Ahmedabad	72 h	●	●	●	◐	·	◐	◐	·	·	·	·	·	·
Air Quality Early Warning System, Delhi (Jena et al.) (2021)	Government operational + paper	India: Delhi-NCR	72 h	●	●	●	●	·	·	·	·	·	·	·	·	·
Decision Support System v1.0, Delhi (Govardhan et al.) (2024)	Government operational + paper	India: Delhi and 19 surrounding districts	120 h	●	●	●	◐	·	·	·	·	◐	●	·	·	·
UrbanEmissions.info India air quality forecasts (ongoing)	Research / NGO operational	India, all districts; Delhi at 1 km	72 h	·	●	●	◐	·	·	·	·	·	·	·	·	◐
CPCB National AQI portal and Sameer app (2014 onward)	Government operational	India, station network	Current conditions	●	·	·	●	·	◐	·	·	·	·	·	·	◐
Graded Response Action Plan (CAQM) (revised 2024)	Policy protocol	India: Delhi-NCR	Days ahead	●	●	·	·	·	·	◐	·	·	◐	·	·	·
US EPA AirNow and the Air Quality Flag Program (ongoing)	Government operational	United States	Next-day	●	●	·	◐	·	●	●	◐	·	·	·	·	◐
UK Daily Air Quality Index (ongoing)	Government operational	United Kingdom	Multi-day	●	●	·	·	·	●	●	·	·	·	·	·	·
Air Quality Health Index, Canada (Stieb et al.) (2008)	Government index + paper	Canada	Hourly index	●	·	·	·	·	◐	·	·	·	·	·	·	·
Hong Kong Air Quality Health Index (ongoing)	Government operational	Hong Kong	Current plus forecast range	●	●	·	●	·	●	◐	·	·	·	·	·	◐

SYSTEM	TYPE	COVERAGE	HORIZON	LIVE	FORE-CAST	WEA-THER	SUB-CITY	UNCER-TAINTY	COHORT ADVICE	ADVICE FROM FORECAST	EXPOSURE WINDOW	EXPLA-NATION	SCEN-ARIO	TRUST / ABSTAIN	UNSEEN CITY	OPEN
European Air Quality Index (EEA) (ongoing)	Government operational	Europe	Current	●	·	·	●	·	●	·	·	·	·	·	·	·
City-specific AQHI for Delhi and Varanasi (George et al.) (2025)	Paper	India: Delhi, Varanasi	Retrospective	·	·	·	·	·	·	·	·	·	·	·	·	·
Forecaster.health (ISGlobal) (2024; air pollution alerts 2025)	Research operational	Europe: 653 regions, 31 countries	Up to 4 days	◐	●	◐	◐	·	●	●	·	·	·	·	·	●
IQAir AirVisual (current)	Commercial product	Global, 500,000+ locations	7 days	●	●	·	◐	·	◐	·	·	·	·	·	·	·
Plume Labs Air Report (2015 onward)	Commercial product	Global urban areas	Hourly, next 24 h	●	●	·	·	·	◐	●	●	·	·	·	·	·
AQI.in (current)	Commercial product	109+ countries, 10,500+ stations	Real-time and history	●	·	·	●	·	◐	·	·	·	·	·	·	·
Ambee Air Quality API (current)	Commercial API	90+ countries	48 h to 7 days (10 days on demand)	●	●	·	●	·	·	·	·	·	·	·	·	·
AirRater (Workman et al. evaluation) (2021)	App + paper	Australia	Current	●	·	·	·	·	●	·	·	·	·	·	·	·
ECMWF CAMS global atmospheric composition (via Open-Meteo) (current)	Operational model + open API	Global	Hourly, 5 days, updated every 12 h	◐	●	●	·	·	·	·	·	·	·	·	·	●
Aurora foundation model (Bodnar et al.) (2025)	Paper + model	Global	5 days for air pollution	·	●	●	·	·	·	·	·	·	·	·	·	◐
AirFormer (Liang et al.) (2023)	Paper	China, 1,085 stations	72 h	·	●	·	●	◐	·	·	·	·	·	·	·	●
PM2.5-GNN (Wang et al.) (2020)	Paper	China	Multi-step	·	●	◐	◐	·	·	·	·	·	·	·	·	◐
E-STGCN (Panja et al.) (2024)	Paper	India: Delhi, 37 stations	Multiple horizons	·	●	·	●	●	·	·	·	·	·	·	·	·
Probabilistic deep learning to quantify uncertainty in air quality forecasting (2021)	Paper	See paper	See paper	·	●	·	·	●	·	·	·	·	·	·	·	·
Conformal prediction of air pollution concentrations, Barcelona Metropolitan Region (n/d)	Paper	Spain: Barcelona	See paper	·	·	·	·	●	·	·	·	·	·	·	·	·

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Cross-city transfer learning for Sentinel-5P NO2 (Janku et al.) (2026)	Paper	Berlin, London, Madrid, Paris, Toronto	Estimation, not forecasting	·	·	●	·	·	·	·	·	·	·	·	●	·
Spatial transfer learning for estimating PM2.5 in data-poor regions (2024)	Paper	See paper	Estimation	·	·	·	·	·	·	·	·	·	·	·	●	·
Ambiguity and novelty rejection for time-series forecasting (Feng et al.) (2025)	Paper	Method	n/a	·	·	·	·	·	·	·	·	·	·	●	·	·
Model-agnostic explainable selective regression via uncertainty estimation (2023)	Paper	Method	n/a	·	·	·	·	·	·	·	·	◐	·	●	·	·
Conformal risk control (Angelopoulos et al.) (2022)	Paper	Method	n/a	·	·	·	·	●	·	·	·	·	·	◐	·	·
WHO AirQ+ (current)	Health-impact tool	Global	Scenario	·	·	·	·	·	·	·	·	·	●	·	·	◐
US EPA BenMAP-CE (current)	Health-impact tool	Global use	Scenario	·	·	·	·	·	·	·	·	·	●	·	·	●
The value of air quality forecasting in the Mid-Atlantic region (see source)	Paper	United States	Next-day	·	●	·	·	·	·	◐	·	·	·	·	·	·
The boy who cried wolf revisited (Roulston and Smith) (2004)	Paper	Method	n/a	·	·	·	·	·	·	◐	·	·	·	·	·	·
Random-forest meteorological normalisation (Grange et al.) (2018)	Paper	Switzerland	Trend analysis	·	·	●	·	·	·	·	·	◐	·	·	·	·
COVID-19 lockdowns cause global air pollution declines (Venter et al.) (2020)	Paper	34 countries	Event study	·	·	●	·	·	·	·	·	·	·	·	·	·
Ambient air pollution and daily mortality in ten cities of India (de Bont et al.) (2024)	Paper	India: ten cities	Retrospective	·	·	·	·	·	·	·	·	·	·	·	·	·
AirSense-X: physics-informed neural and explainable AQI prediction (2025)	Paper	India, multiple cities	See paper	·	·	·	·	·	·	·	·	●	·	·	·	·
Forecasting and validating Delhi's AQI: a multi-model approach (2026)	Paper	India: Delhi, one station	Next-day	·	●	·	·	·	·	·	·	●	·	·	·	·

SYSTEM	TYPE	COVERAGE	HORIZON	LIVE	FORE-CAST	WEA-THER	SUB-CITY	UNCER-TAINTY	COHORT ADVICE	ADVICE FROM FORECAST	EXPOSURE WINDOW	EXPLA-NATION	SCEN-ARIO	TRUST / ABSTAIN	UNSEEN CITY	OPEN
Dual-branch explainable AI framework for AQI across Indian cities (2026)	Preprint	India, major cities	See preprint	·	◐	·	·	·	·	·	·	●	·	·	·	·
Student and open projects on the same Kaggle dataset (2020-2026)	Open-source projects	India	Mostly same-day	·	·	·	·	·	·	·	·	◐	·	·	·	●
Instructor-worker LLM system for air-quality policy recommendation (2025)	Preprint	United States: Los Angeles	Retrospective	·	·	·	·	·	◐	·	·	◐	·	·	·	·
ESoWC air-quality bias correction of CAMS (see repository)	Open-source project	Station sites	Forecast post-processing	·	●	●	●	·	·	·	·	·	·	·	·	●
Machine learning for urban air quality analytics: a survey (2023)	Survey	Global literature	n/a	·	·	·	·	·	·	·	·	·	·	·	·	·
CPCB continuous monitoring data for India, 2015-2019 (Princeton Data Commons) (n/d)	Open dataset	India, all continuous stations	Historical	·	·	·	●	·	·	·	·	·	·	·	·	●
AirGuard India 1.0 (this project, as deployed) (2026)	Student project, deployed	India, 12 cities	Next-day, replayed on historical dates	·	●	·	·	·	●	●	◐	●	●	◐	·	●
Systems documenting the capability (of 48)				19	24	13	19	5	15	10	3	9	5	4	2	15

B.3 What each system leaves open

SYSTEM	WHAT IT PROVIDES	WHAT IT LEAVES OPEN
Google Maps Platform Air Quality API	Not disclosed. Recommendations for the general public and for children, elderly, pregnant women, athletes, and people with asthma or heart conditions	No uncertainty or confidence documented for forecasts; data sources and method undisclosed; recommendations are text keyed to index level; no stated guarantee on advisory error; closed.
Google Air View+ (India)	Multi-layer AI fusion. Follows National AQI categorisation and guidance	Nowcast rather than next-day forecast in the sources reviewed; no cohort-specific guidance, uncertainty or explanation documented; closed.
SAFAR-India (IITM / IMD)	Chemistry-transport modelling. Health advisory and precautions keyed to AQI	Four cities; advisory is generic by AQI level rather than by cohort; no uncertainty communicated.
Air Quality Early Warning System, Delhi (Jena et al.)	WRF-Chem with chemical data assimilation	Delhi only; computationally heavy physical model; no personalised guidance; no uncertainty band for the public.
Decision Support System v1.0, Delhi (Govardhan et al.)	WRF-Chem source apportionment	Delhi only; scenarios are sector emission interventions, not translated into who is affected; no cohort or uncertainty layer.
UrbanEmissions.info India air quality forecasts	WRF + CAMx chemical transport	Physical-model forecast without station-level statistical correction, cohort guidance or uncertainty in the sources reviewed.
CPCB National AQI portal and Sameer app	Index computed from measurements. Health statements attached to each AQI category	Describes the present, not tomorrow; one message for everyone in a category; the maximum sub-index discards the rest of the mixture.
Graded Response Action Plan (CAQM)	n/a (decision protocol). Population-wide restrictions by stage	Triggers on a single regional AQI threshold; no differentiation by vulnerable group; forecast uncertainty is not part of the stated trigger.
US EPA AirNow and the Air Quality Flag Program	Agency forecasts. Guidance for sensitive groups (lung disease, older adults, children, people active outdoors); school flag colours drive activity changes	Two tiers only (everyone / sensitive groups); timing guidance is generic; no uncertainty shown.

SYSTEM	WHAT IT PROVIDES	WHAT IT LEAVES OPEN
UK Daily Air Quality Index	Operational forecast. Separate advice for the general population and for at-risk individuals (lung problems, heart problems, older people)	Two tiers; advice keyed to band rather than to the pollutant mixture; no uncertainty shown.
Air Quality Health Index, Canada (Stieb et al.)	Sum of excess mortality risk across pollutants, scaled 0-10. Index is itself health-derived	Coefficients are Canadian; no cohort-specific weights; a framework to follow rather than a competitor.
Hong Kong Air Quality Health Index	Health-risk based index. Precautionary advice for children, the elderly and people with heart or respiratory illness, separately from the general public	Two tiers; no uncertainty; no scenario analysis.
European Air Quality Index (EEA)	Index. Health messages for the general population and for sensitive groups	Two tiers; not forecast-driven in the source reviewed.
City-specific AQHI for Delhi and Varanasi (George et al.)	Generalised additive model, quasi-Poisson. Excess mortality risk per AQI category (2% moderate to 16% severe)	Two cities; retrospective; no forecast, no cohort weights, no application. Shows a pooled index can misrepresent local risk.
Forecaster.health (ISGlobal)	Epidemiological exposure-response models applied to forecasts. Mortality-risk alerts by sex and age group for PM2.5, PM10, NO2 and O3	Europe only; groups are sex and age, not condition or occupation; no decision guarantee on warnings or scenario analysis documented. The closest conceptual prior art, and epidemiologically stronger than a hand-weighted index.
IQAir AirVisual	Described only as machine learning and AI. Health recommendations	Method and accuracy undisclosed; no uncertainty; generic recommendations.
Plume Labs Air Report	Described as environmental AI. Advice tailored to runners, cyclists, parents and outdoor dining	Activity profiles rather than health-vulnerability cohorts; 24-h horizon; no uncertainty; method undisclosed.
AQI.in	n/a. Personalised health recommendations by location	No next-day forecast documented in the listing reviewed; no uncertainty or explanation.
Ambee Air Quality API	Not disclosed	Vendor-reported accuracy; no uncertainty, cohort guidance or explanation documented; closed.

SYSTEM	WHAT IT PROVIDES	WHAT IT LEAVES OPEN
AirRater (Workman et al. evaluation)	n/a. Personal symptom tracking and alerts	No forecast; evidence is qualitative. Useful as evidence that people with existing conditions act on such information.
ECMWF CAMS global atmospheric composition (via Open-Meteo)	Physical model	About 45 km grid cannot resolve a city's hotspots; modelled, not measured. The benchmark any Indian statistical forecast must beat.
Aurora foundation model (Bodnar et al.)	Foundation model, fine-tuned per task	Grid-scale concentrations; no health layer, no decision layer; heavy to run.
AirFormer (Liang et al.)	Transformer with a stochastic latent stage	Needs a dense station graph; no health layer; uncertainty is modelled internally rather than calibrated for users.
PM2.5-GNN (Wang et al.)	Knowledge-enhanced graph neural network	Requires a multi-city graph and meteorology; no uncertainty or health layer.
E-STGCN (Panja et al.)	Graph convolution with extreme-value-guided LSTM; conformal prediction intervals	One city; no health or decision layer; no unseen-city test. The closest Indian work on extremes with conformal intervals.
Probabilistic deep learning to quantify uncertainty in air quality forecasting	Probabilistic deep learning	Uncertainty reported as a modelling result, not carried into a public advisory.
Conformal prediction of air pollution concentrations, Barcelona Metropolitan Region	Conformal prediction	Intervals without a decision or advisory layer.
Cross-city transfer learning for Sentinel-5P NO2 (Janku et al.)	CatBoost, leave-one-city-out	Estimates concentrations rather than forecasting; five wealthy cities; shows leave-one-city-out is already an established protocol.
Spatial transfer learning for estimating PM2.5 in data-poor regions	Transfer learning	Estimation at unseen locations rather than next-day advisory.
Ambiguity and novelty rejection for time-series forecasting (Feng et al.)	Error-variance ambiguity rejection plus VAE / Mahalanobis novelty rejection	General method; does not examine which cases are rejected in a safety-critical advisory setting.
Model-agnostic explainable selective regression via uncertainty estimation	Selective regression	General method, not air quality.

SYSTEM	WHAT IT PROVIDES	WHAT IT LEAVES OPEN
Conformal risk control (Angelopoulos et al.)	Distribution-free control of an expected loss	Guarantee assumes exchangeability, which a drifting pollution series violates; needs adaptation for time series.
WHO AirQ+	Epidemiological exposure-response functions. Attributable cases, years of life lost	Desktop assessment tool, not a forecast or advisory system.
US EPA BenMAP-CE	Exposure-response functions with built-in percentage roll-back of concentrations. Mortality and morbidity avoided, with valuation	Assessment tool; annual-scale outcomes; not connected to a daily forecast or advisory. Direct prior art for percentage-reduction scenarios.
The value of air quality forecasting in the Mid-Atlantic region	Cost-loss decision model. Protective-action framing	One cost-loss ratio for all users; not cohort-specific; no distribution-free guarantee.
The boy who cried wolf revisited (Roulston and Smith)	Cost-loss with false-alarm intolerance	Theory; motivates measuring over-warning alongside under-warning.
Random-forest meteorological normalisation (Grange et al.)	Random forest	Explains why an uncorrected before/after comparison cannot be read causally.
COVID-19 lockdowns cause global air pollution declines (Venter et al.)	Weather-adjusted comparison. Health-burden implications	The reference method for a lockdown effect; the present project's -34% is not weather-adjusted.
Ambient air pollution and daily mortality in ten cities of India (de Bont et al.)	Poisson time series with instrumental-variable causal analysis. 1.4% higher daily mortality per 10 ug/m3 (2-day mean); 3.6% in the causal analysis	All-age mortality only; the Indian evidence base an advisory index should be anchored to.
AirSense-X: physics-informed neural and explainable AQI prediction	Physics-informed neural network with XAI	Headline accuracy near the level a same-day formulation produces by construction; no persistence baseline, uncertainty or unseen-city test in the summary reviewed.
Forecasting and validating Delhi's AQI: a multi-model approach	Linear, random forest, XGBoost with SHAP	Single station; no weather; no uncertainty. An honest next-day result that shows how hard the task is without lag-rich features.
Dual-branch explainable AI framework for AQI across Indian cities	XGBoost, LightGBM, CatBoost, linear; LSTM, GRU, TCN, Transformer; multi-level XAI	Model comparison with explanation; no health, uncertainty or decision layer in the summary reviewed.

SYSTEM	WHAT IT PROVIDES	WHAT IT LEAVES OPEN
Student and open projects on the same Kaggle dataset	Random forest, KNN, gradient boosting, LSTM. None or generic	Commonly predict the same day's AQI from the pollutants that define it, with random splits and no persistence baseline.
Instructor-worker LLM system for air-quality policy recommendation	Multi-agent large language model. LLM-generated daily health advisories compared against issued ones	Generative advice without a calibrated forecast or an error guarantee.
ESoWC air-quality bias correction of CAMS	Machine- and deep-learning bias correction	Establishes statistical post-processing of CAMS as a known technique; not India-specific.
Machine learning for urban air quality analytics: a survey	Review of temporal, spatiotemporal and attention models	Confirms that forecasting architecture alone is a crowded field.
CPCB continuous monitoring data for India, 2015-2019 (Princeton Data Commons)	n/a	Ends 2019; a station-level complement to the city-level file used today.
AirGuard India 1.0 (this project, as deployed)	LightGBM regression; LightGBM + SMOTE classification as deployed. Seven-cohort Health Advisory Index with hand-set weights	Not live; no weather; no interval; deployed confidence is miscalibrated; index saturates on bad days; advisory weights unvalidated.

B.4 What the corpus shows

1. **Every individual feature AirGuard has is already established.** Next-day forecasting (23 of 47), cohort or sensitive-group advice (14), sub-city resolution (19), explanation (8) and scenario analysis (4) all exist in operational systems. None of these may be claimed as novel.
2. **India already has operational physical-model forecasting** that a tabular model on city-day data will not out-resolve: 400 m for Delhi (AQEWS), district-level nationwide (UrbanEmissions), 72 hours in four cities (SAFAR), and a decision-support system that attributes Delhi's pollution to sectors and districts.
3. **Cohort-differentiated advice is standard practice** — the UK, Hong Kong, the EEA and the US EPA publish separate guidance for at-risk groups, and Google's API returns recommendations for six groups in over 100 countries. AirGuard's seven cohorts are an extension in granularity, not a new idea.
4. **The closest conceptual prior art is Forecaster.health** (ISGlobal): operational, forecast-driven mortality-risk alerts by sex and age for 653 European regions, built on epidemiological models. It is methodologically stronger than a hand-weighted index. It does not cover India, does not differentiate by condition or occupation, and documents no error guarantee on its alerts.

5. **Health-based indices exist and one has been derived for India.** Canada's AQHI sums excess mortality risk across pollutants; George et al. (2025) derived city-specific versions for Delhi and Varanasi and showed a pooled index misstates local risk. An index that claims to reflect health must now be measured against this standard.
6. **Uncertainty lives in research papers, not in advisories.** Five sources document uncertainty; all are methods papers. **None of the 47 documents forecast, cohort advice and uncertainty together, and none connects uncertainty to the advice it issues.**
7. **Percentage-reduction scenarios are a twenty-year-old technique** (BenMAP's roll-back), and BenMAP and AirQ+ translate them into deaths and illness avoided, which is more than counting advisory-days. None of the 47 links a scenario to cohort-level advisory burden, which is a reporting difference rather than a methodological one.
8. **Leave-one-city-out is an established protocol** (Janku et al. 2026), and the pilot shows the effect is small here.
9. **Published Indian AQI-prediction work frequently reports near-perfect accuracy** from same-day formulations and random splits. A strictly chronological, baseline-referenced evaluation is a real point of difference, but it is rigour, not novelty.

C. Closest-system comparison

● yes, documented · ◐ partly · · not documented in the source reviewed. "2.0 target" is a design goal, **not an existing capability**.

#	QUESTION	GOOGLE AQ API	FORECASTER. HEALTH	SAFAR	AQEWS / DSS DELHI	URBAN EMISSIONS	IQAIR	PLUME	E-STGCN	AIRGUARD 1.0 (TODAY)	AIRGUARD 2.0 TARGET
1	Data used	undisclosed	AQ forecasts + mortality	stations + model	WRF-Chem + assimilation	WRF + CAMx	stations + sensors	sensors + model	37 Delhi stations	CPCB city-day 2015–20	stations + CAMS + forecast weather
2	Live data	●	◐	●	●	·	●	●	·	**no**	Proposed
3	Predicts tomorrow's AQI	●	◐ risk, not AQI	●	●	●	●	◐ 24 h	●	● replay only	Proposed
4	Forecasts individual pollutants	◐	◐	◐	●	●	·	·	●	**no**	Proposed
5	Uses weather forecasts	·	◐	●	●	●	·	·	·	**no**	Proposed
6	Station-level or finer	● 500 m	◐ region	◐	● 400 m	◐ district	◐	·	●	**no**	Proposed
7	Multiple cities	●	●	● 4	Delhi	●	●	●	Delhi	● 12	●
8	Tested on unseen cities	·	·	·	·	·	·	·	·	pilot: +3.5% MAE	Proposed
9	Quantifies uncertainty	·	·	·	·	·	·	·	●	**no**	Proposed
10	Detects out-of-distribution input	·	·	·	·	·	·	·	·	◐ z-score	Proposed
11	Detects model drift	·	·	·	·	·	·	·	·	**no**	Proposed
12	Abstains when unreliable	·	·	·	·	·	·	·	·	**no**	data-integrity only
13	Personalised guidance	● 6 groups	● sex, age	◐ generic	·	·	◐	◐ activities	·	● 7 cohorts	●
14	Guidance driven by the forecast	·	●	◐	·	·	·	●	·	●	●
15	Cohort-specific exposure advice	●	◐	·	·	·	·	◐	·	● unvalidated	Proposed, anchored
16	Lower-exposure window	·	·	·	·	·	·	●	·	◐	Proposed

#	QUESTION	GOOGLE AQ API	FORECASTER. HEALTH	SAFAR	AQEWS / DSS DELHI	URBAN EMISSIONS	IQAIR	PLUME	E-STGCN	AIRGUARD 1.0 (TODAY)	AIRGUARD 2.0 TARGET
17	Window from history or forecast	—	—	—	—	—	—	forecast	—	history	forecast
18	Explainable AI	·	·	·	◐ source shares	·	·	·	·	◐ fixed example	per forecast
19	Scenario or causal analysis	·	·	·	● causal, by sector	·	·	·	·	● scenario only	scenario + sensitivity
20	Scenario linked to vulnerable groups	·	·	·	·	·	·	·	·	● advisory-days	●
21	Research-grade diagnostics exposed	·	·	·	◐ papers	·	·	·	◐ paper	●	●
22	Data and model provenance exposed	·	◐	·	◐	◐	·	·	◐	●	●
23	Externally validated	·	◐	·	● peer-reviewed	·	·	·	◐ preprint	**no**	Future research
24	Reproducible	·	◐ open access	·	·	·	·	·	◐	● one command	●
25	Nature	product	research operational	government	government	NGO	product	product	paper	student project	research platform

D. Gap-to-solution matrix

Every major 2.0 feature answers a row here. A feature that answers no row is not built.

EXISTING LIMITATION	EVIDENCE	WHY IT MATTERS	AIRGUARD 2.0 RESPONSE	EXPERIMENT	SUCCESS METRIC
Advisories are issued from a point forecast with no stated error rate	0 of 47 sources link uncertainty to advice; pilot P5: 18.0% of polluted days under-warned	A missed warning harms the people the advisory exists for	Issue each cohort's advisory from a quantile chosen to hold under-warning below a declared target	Risk-controlled advisory on rolling-origin data	Realised under-warning ≤ target on unseen future data, with fewer over-warnings than a uniform one-band shift
Intervals fail on extreme days	Pilot P2: 57–75% coverage on *Severe* days at 90% nominal	Average-case calibration hides the worst case	Regime-aware conformal intervals with weather features	Conditional coverage by band and season	≥ 85% coverage on *Poor*-or-worse days at 90% nominal
Forecast confidence shown to users is uncalibrated	Pilot P1: ECE 0.093 on the live classifier	A wrong "99%" destroys trust	One predictive distribution; band probabilities derived from it	Reliability diagram on future data	ECE ≤ 0.03
Systems always answer	0 of 23 forecasting sources document abstention; pilot P4 shows naive abstention is unsafe	Silence when air is worst is worse than a cautious answer	Withhold only on data-integrity failure; otherwise widen and escalate	Compare three policies on declined-day composition	No *Severe* day declined for uncertainty alone
No weather in the model	Audit; *Severe* bias −22.6	Dispersion drives next-day change	Day-ahead forecast meteorology, archived at true lead time	Ablation: pollution → + weather → + CAMS → + fire	MAE and *Severe*-day bias, each with a bootstrap interval

EXISTING LIMITATION	EVIDENCE	WHY IT MATTERS	AIRGUARD 2.0 RESPONSE	EXPERIMENT	SUCCESS METRIC
Statistical forecasts are rarely compared with the free physical forecast	CAMS is public and global	Beating persistence is not the bar a user faces	CAMS as a baseline and as an input	Station model vs raw CAMS vs CAMS post-processed	Skill score against CAMS
Cohort weights are hand-set	Audit; George et al. 2025 and de Bont et al. 2024 provide Indian evidence	An index that claims health relevance must trace to evidence	Anchor to published guidance tiers and Indian concentration–response functions; publish sensitivity	Advisory flips under ±20% weight perturbation	Share of advisories stable under perturbation
Index saturates	Audit: six cohorts at 100.0 on one day	No discrimination on the worst days	Unbounded scale with labelled bands	Distribution of the index on *Very Poor* and *Severe* days	No more than 1% of values at any cap
Lowest-exposure window is a climatology	Audit	Same answer every day of a season	Hourly forecast for tomorrow, with the absolute level stated	Hit rate of the recommended window against the observed minimum	Better than the climatological window on held-out days
Single chronological holdout	Audit: CV MAE 29.9 vs test 15.1	One easy test year flatters the model	Rolling-origin evaluation across years	Per-origin error	Report the distribution, not one number
Not live	Audit	The product claim depends on it	Ingestion with provenance and quality flags	Seven days of unattended ingestion	≥ 95% of scheduled runs succeed; staleness always shown
Scenario results carry no uncertainty	Audit	A single number overstates precision	Sensitivity bands; weather-normalised baseline	Scenario repeated across years and seasons	Range reported with every scenario

D.1 Feature novelty classification

AIRGUARD FEATURE	CLASS	REASON
Next-day AQI forecasting	Existing / common	23 of 47 sources
Gradient-boosted trees, SHAP	Existing / common	Standard
Live dashboard, map	Existing / common	Standard
Cohort-specific advice	Improved implementation	Tiers exist; seven cohorts is finer granularity
Lowest-exposure window from a forecast	Improved implementation	Plume Labs did this for 24 hours
Percentage-reduction scenarios	Existing / common	BenMAP roll-back
Scenario → cohort advisory-days	Integration contribution	Not found among the 47; a reporting choice
Forecast + cohort advice + calibrated uncertainty in one pipeline	Integration contribution	Co-occurrence not found among the 47
Chronological evaluation, leakage demonstration, persistence and CAMS baselines for Indian cities	Experimental contribution	Counters a documented weakness in the Indian literature
Leave-one-city-out and excluded-city tests	Experimental contribution	Protocol exists; this application does not appear among the 47
Evidence that uncertainty-based abstention silences severe days	Experimental contribution	Pilot P4; to be confirmed on new data
Risk-controlled cohort advisories	**Methodological contribution (candidate)**	Section E
Unified citizen / researcher / policymaker interface	Product contribution	Not a research claim

E. Research hypothesis

Selected because it is the one gap where the corpus is empty, the pilot shows a large measurable effect, and an existing method (conformal risk control) offers a principled route.

> **H1 (primary).** For next-day air-quality advisories in Indian cities, an advisory issued from a quantile of a calibrated predictive distribution, with the quantile set per cohort by conformal risk control, holds the rate of under-warning on polluted days at or below a pre-declared cohort-specific target on unseen future data, and does so with significantly fewer over-warnings than (a) a point forecast shifted up by one band and (b) a fixed quantile chosen without calibration.

Supporting hypotheses, each a separate experiment:

- **H2 — Meteorology.** Day-ahead forecast meteorology, used strictly at its true lead time, reduces MAE and specifically reduces the under-prediction on *Poor*-or-worse days, relative to the pollution-only model.
- **H3 — Conditional coverage.** Regime-aware conformal intervals reach at least 85% coverage on *Poor*-or-worse days at 90% nominal, where the three pilot methods reach 57–75% on *Severe* days.
- **H4 — Beyond the free forecast.** A station-level statistical model that takes CAMS as an input beats both raw CAMS and the pollution-only model.
- **H5 — Transfer.** H1's guarantee continues to hold in cities absent from training and in the cities the original completeness gates excluded.

Every hypothesis is falsifiable and each has a pre-declared metric in the matrix above. A negative result on any of them is reported.

F. Novelty statement

What may be said, on the evidence so far:

> *We propose a framework in which next-day air-quality advisories for vulnerability cohorts are issued from a calibrated predictive distribution at a cohort-specific risk level, so that the rate of under-warning is controlled rather than incidental. Among the 47 operational systems, products and papers we reviewed, none documents forecast-driven cohort advice together with calibrated uncertainty, and none attaches a stated error rate to the advice it issues.*

What may **not** be said:

- That AirGuard is the first, or the only, system to do any single thing in the feature list. Each is prior art.
- That the Health Advisory Index is validated, clinical, or evidence-derived, until Part 2 section L is carried out.
- That the simulator predicts the effect of any intervention.
- That the risk-controlled advisory is new to the literature. A review of 47 sources does not establish that. A dedicated search of conformal decision-making, cost-loss forecasting and impact-based warning literature is a precondition for the word "novel", and is scheduled as the first task of the paper phase.

Part 2 — Research design

How the hypotheses in Part 1 will be tested. Everything in this part is **Proposed** unless marked otherwise. Each choice states why it was made, what was considered instead, and what evidence would confirm or overturn it.

Each element is classified: **ER** essential for research · **EP** essential for product · **HV** high-value optional · **FW** future work.

H. Dataset strategy

H.1 Sources, and what was actually verified

Access was tested from the development machine on 7 October 2026. "Verified" means a real request returned real data on that date.

SOURCE	ROLE	ACCESS	STATUS ON 7 OCT 2026	CONSTRAINT TO RESPECT
CPCB real-time AQI on data.gov.in	Official live station measurements	Free API key from the portal	**Not verified.** `api.data.gov.in` refused the connection; the resource page blocks automated reading. One search result reported the API as not currently offered.	Must be confirmed with the team's own key before the design depends on it
CPCB CCR portal	Official station history	Manual download, limited span per request	Documented; no API found	Backfill only; cannot feed a scheduler
OpenAQ v3	Aggregated station measurements	Free API key	Endpoint reachable; returns 401 without a key	Label as *OpenAQ, originating from CPCB* — never as an official CPCB feed. India freshness to be checked with a key

SOURCE	ROLE	ACCESS	STATUS ON 7 OCT 2026	CONSTRAINT TO RESPECT
CAMS global, via Open-Meteo Air Quality API	Modelled pollutant analysis and 5-day hourly forecast	No key	**Verified.** Returned current PM2.5, PM10, NO2, SO2, O3, CO for Chennai and Delhi	Modelled on a ~45 km grid, not measured. Attribution to CAMS and Open-Meteo required. History from August 2022
Open-Meteo Forecast API	Day-ahead meteorology	No key	**Verified.** Temperature, humidity, wind speed and direction, pressure, precipitation, **boundary-layer height**, radiation	Free tier is non-commercial; any paid offering needs their commercial tier or self-hosting
Open-Meteo Previous Runs API	Archived forecasts *at their original lead time*	No key	**Verified.** Returned the forecast made one and two days earlier for a past date	Starts January 2024. This is what makes leakage-free training on forecast weather possible
Open-Meteo Historical Weather (ERA5)	Reanalysis for 2015–2020	No key	Documented, not probed	Reanalysis is not a forecast; see H.3
NASA FIRMS	Active fires (VIIRS 375 m)	Free MAP_KEY	Documented, not probed	5,000 transactions per 10 minutes
Sentinel-5P / MODIS AOD	Satellite columns	Earth Engine or Copernicus	Not probed	Cloud gaps, one overpass a day; kept only if the ablation earns it
India Meteorological Department	Official weather	—	No openly documented API was found in this review	Not used. Open-Meteo is therefore labelled a third-party source
Traffic	Congestion proxy	—	No legally clean free source identified	**FW**
Kaggle city-day 2015–2020 (in use)	Historical baseline	CC0	In the repository	Ends July 2020; city-level
Princeton CPCB 2015–2019	Station-hour history	Open	Located, not downloaded	Station-level complement to the file above

Why Open-Meteo for weather, given the brief asks for official sources first. Considered: IMD (no documented open API found), direct ECMWF or NOAA GFS access (free but requires GRIB handling and an archive the project would have to build itself). Open-Meteo redistributes those same national models, and uniquely offers forecasts archived at lead time, which no other free source reviewed provides. **Evidence that would overturn this:** an IMD API with archived forecasts, or measured skill for Indian cities that is clearly worse than an alternative.

H.2 Provenance contract — EP

Every stored measurement carries: `source`, `source_station_id`, `station_name`, `city`, `latitude`, `longitude`, `pollutant`, `value`, `unit`, `averaging_period`, `observed_at` (UTC), `retrieved_at` (UTC), `quality_flag`, and `is_modelled`. The interface shows a measured value and a modelled value differently and never merges them silently. When no fresh measurement exists, the interface says **"Data temporarily unavailable"** and may show the modelled estimate, labelled as such.

H.3 Two training tracks, kept separate — ER

The historical file and the live sources do not overlap in time, and they cannot be mixed without leaking information.

	TRACK A — HISTORICAL	TRACK B — OPERATIONAL
Period	2015 – mid-2020	January 2024 – present
Pollution	Kaggle city-day; Princeton station-hour	Station measurements via OpenAQ or CPCB
Weather	ERA5 reanalysis	Forecasts archived at one-day lead
CAMS	Not available	Forecast archive from August 2022
Use	Method development, long-run evaluation, the 2015–2020 pilot	**The model that is deployed**
What it can claim	An **upper bound** on the value of weather (a "perfect-prognosis" experiment)	Real skill under real forecast error

Reanalysis describes the weather that happened. Feeding tomorrow's reanalysis to a model predicts tomorrow with knowledge no forecaster has. Track A weather results are therefore always labelled an upper bound, and no Track A weather model is ever served.

The period from July 2020 to the end of 2023 is backfilled with pollution-only data where the station archive allows, and is used to test stability across years.

H.4 The prediction-time contract — ER

A forecast for day *D + 1* is issued at a fixed time on day *D* (18:00 IST proposed). A feature may use an observation only if its `observed_at` precedes the issue time by the source's publication delay, and a forecast variable only if the model run was available by then. At issue time the full feature vector is written to a snapshot table and is never recomputed. Training reads snapshots, so the model can only ever be trained on what it would actually have seen.

I. Preprocessing pipeline — ER

Retained from 1.0 without change: chronological split; statistics fitted on training data only; event-aware outlier rule; causal imputation with a stated cost; the leakage demonstration.

Added for 2.0:

1. **Unit and time normalisation.** All timestamps to UTC with the IST calendar day as the target period; CO in mg/m³, everything else in µg/m³, converted at ingestion with the conversion recorded.
2. **Station quality gates.** Flat-line detection (identical values for six or more hours), range checks against the existing plausibility ceilings, spike checks against neighbouring stations, and a completeness rule for daily aggregates (a minimum number of valid hours per day, set from the CPCB index method document and recorded in configuration).
3. **Staleness.** A station with no reading for three hours is marked stale; a stale station contributes no feature and triggers the integrity path in section N.
4. **Duplicates.** A uniqueness constraint on (source, station, pollutant, observed_at, averaging_period); ingestion is idempotent.
5. **Targets computed, not downloaded.** The next-day AQI is recomputed from concentrations with the CPCB breakpoints, as 1.0 already does, so the label is identical across sources.
6. **Feature selection stays inside each training fold.** In 1.0 selection ran once on the training period; in rolling-origin evaluation it is repeated per origin.
7. **Event flags are kept by rule, not by selection.** They were discarded in 1.0 by a statistical filter; rare events have low marginal importance and high importance on the days they occur.

J. Model benchmark plan — ER

The current LightGBM is the baseline to beat, not the presumed winner.

TIER	MODELS	ADMITTED WHEN
0 — References	Persistence; 7-day seasonal persistence; city-and-day-of-year climatology; **raw CAMS forecast**	Always

TIER	MODELS	ADMITTED WHEN
1 — Linear	Ridge; SARIMAX with exogenous terms	Always; SARIMAX only if residual autocorrelation warrants it
2 — Trees	LightGBM (current), XGBoost, CatBoost, random forest; quantile variants of each	Always
3 — Sequence	GRU, temporal convolutional network, PatchTST, Temporal Fusion Transformer	Only with station-hour data on the order of a million rows
4 — Spatiotemporal	Graph networks with stations as nodes and edges from distance and forecast wind direction	Only with roughly a hundred or more stations each having two years of history

Adoption rule. A model from a higher tier replaces the best lower-tier model only if all three hold: its rolling-origin MAE is lower by more than the bootstrap confidence interval (and a Diebold–Mariano test rejects equal accuracy); its bias on *Poor*-or-worse days is not worse; and it fits the serving budget. Otherwise the simpler model is kept and the comparison is reported as a negative result.

Why this, and not "use a transformer". The corpus contains many deep spatiotemporal models (AirFormer, PM2.5-GNN, E-STGCN) trained on hundreds to thousands of stations. On twelve city series the pilot already shows a pooled tree model transferring across cities with a 3.5% loss. Depth will be earned by data volume or not used.

Targets. Primary: next-day AQI per station and per city. Also evaluated: next-day concentration per pollutant, with AQI and dominant pollutant *computed* from the predicted concentrations, compared against predicting AQI directly. Hourly profile for the next day, for the exposure window. Days two and three are produced but **shown only if** their skill against persistence is positive with its confidence interval and their interval coverage holds.

K. Evaluation plan — ER

K.1 Protocols

PROTOCOL	QUESTION IT ANSWERS
Rolling origin, monthly refits, expanding window	Skill on the future, across many test periods rather than one
Strict final holdout (most recent winter, untouched until the end)	The number that goes in the abstract
Leave-one-city-out	Transfer to a city absent from training
Excluded-city test	Transfer to the cities the completeness gates removed

PROTOCOL	QUESTION IT ANSWERS
Station holdout within a city	Transfer to a new monitor
Regime slices	Error by AQI band, season, festival window, and extreme episodes

No random split is used anywhere. No synthetic sample ever enters a validation or test fold.

K.2 Metrics

ASPECT	METRICS
Point forecast	MAE (primary), RMSE, bias; skill score against persistence and against CAMS; all reported by band, city, season
Extremes	MAE and bias on *Poor*-or-worse days; hit rate and false-alarm ratio for AQI > 300
Category	Macro-F1, per-class recall with *Severe* recall stated, ordinal error
Probability	Expected calibration error, Brier score, reliability diagram
Intervals	Coverage and mean width, overall and **conditional** on band, city and season; interval score
Advisory	Under-warning rate on polluted days; over-warning rate; both per cohort
Uncertainty of the metrics	Block bootstrap confidence intervals on every headline number

Severe days are few (44 in the 1.0 test set). Every claim about them carries its sample size and interval, and conclusions are not drawn from a difference inside that interval.

K.3 Ablation — ER

Components are added one at a time and kept only if they pay:

```
persistence → + lag and rolling features → + calendar and event flags → + forecast meteorology
→ + CAMS forecast → + fire counts upwind → + neighbouring stations → + satellite columns
```

Each step reports the change in MAE, in *Poor*-or-worse bias and in interval width, with a confidence interval. A step whose gain lies inside the interval is dropped and reported as such.

Uncertainty and the risk-controlled advisory — ER

One predictive distribution

1.0 runs a regressor and a separate classifier that can disagree. 2.0 fits quantile models and conformalises them, giving a full predictive distribution for next-day AQI. Everything shown is read from it: the expected value, the likely range, and the probability of each band (the mass of the distribution falling inside that band's limits). The band probabilities and the number can no longer contradict each other.

Candidates for the conformal layer: conformalised quantile regression (pilot: 90.2% overall, 75% on *Severe*); Mondrian conditioning on predicted band and season; adaptive conformal inference, which adjusts the level online as the series drifts. **Chosen by:** conditional coverage on *Poor*-or-worse days and interval score, on rolling origins.

The advisory as a controlled decision

For cohort *c*, choose in advance the largest acceptable rate of under-warning, α_c. The advisory band is read from quantile *q* of the predictive distribution, and *q* is set on a calibration window as the smallest value whose under-warning rate there does not exceed α_c (conformal risk control). Because the loss is monotone in *q*, the procedure has a finite-sample guarantee under exchangeability; because pollution series drift, the calibration window rolls forward and the guarantee is checked empirically rather than assumed.

Consequences:

- Vulnerability enters twice and separably: in how the mixture is scored (section L) and in how much risk is tolerated. A cardiac patient's advisory is issued further up the distribution than a healthy adult's.
- The cost is visible. Every setting reports its over-warning rate beside its under-warning rate, as in pilot P5.
- The α values are **value judgements, not statistics**. They are proposed with a public-health reviewer, published, and never presented as derived.

Alternatives considered. Shifting every vulnerable cohort up one band (what official tiers effectively do): simple, used as the main baseline, expected to over-warn. Abstaining under uncertainty: rejected on pilot P4. A fixed high quantile for everyone: a second baseline. **Evidence that would overturn the choice:** if the controlled rule does not beat the one-band shift on over-warning at matched under-warning, the simpler rule is adopted and that is the finding.

What a user sees

Expected AQI 187 · likely range 165–211 · most likely "Moderate" (62%), "Poor" (31%). A confidence word (high, moderate, low) is defined by the interval's width relative to the band it sits in, with the definition published. The phrase "low-confidence forecast — interpret cautiously" appears when the range spans three or more bands.

L. Health Advisory Index methodology — ER / FW

The index remains **informational decision support**. It is not a clinical score and is not described as one anywhere.

L.1 What changes

ISSUE IN 1.0	CHANGE IN 2.0
Ceiling at 100 makes six cohorts identical on bad days	Unbounded index with labelled bands; saturation reported as a metric
Weights hand-set	Two-layer design below
No evidence of robustness	Published sensitivity analysis
Text not reviewed	Advisory text and translations reviewed by a clinician or public-health professional before release

L.2 Two layers

Layer 1 — tiers that authorities already publish (EP). CPCB attaches health statements to each AQI category; the UK, Hong Kong, the EEA and the US EPA publish separate guidance for at-risk groups, which in effect starts their advice one category earlier. Each AirGuard cohort is mapped to these published tiers, with the source cited beside every sentence of advice. This layer makes no new scientific claim.

Layer 2 — pollutant-specific emphasis (ER, exploratory). The finer differences between cohorts (for example, more weight on SO2 and O3 for asthma) remain a constructed prototype. They are shown as such, and are tested rather than asserted:

- *Sensitivity.* Each factor is perturbed by ±20% and the co-exposure coefficient across 0 to 0.5. The reported quantity is the share of city-days whose advisory band changes. If advice is unstable under plausible perturbation, the layer is simplified.
- *Anchoring.* An alternative index is built in the AQHI form — the sum of excess risk per pollutant — using Indian concentration–response estimates (George et al. 2025 for Delhi and Varanasi; de Bont et al. 2024 for ten cities). The two indices are compared day by day. Where they disagree is where the hand-set weights need justification.

L.3 What would count as validation — FW

Agreement with health outcomes: daily respiratory and cardiovascular admissions or emergency visits by age group, from a hospital or health-department partner. That data is not available to the project. Until it is, the index is described as *evidence-informed and unvalidated*, and Forecaster.health is cited as the system that has done this properly for Europe.

L.4 Privacy

Users pick a broad cohort; no diagnosis, name or account is required. The selection is kept on the device.

Lowest-exposure window 2.0 — EP

Tomorrow's hourly profile is forecast from the CAMS hourly shape, corrected by each station's own learned diurnal pattern and by forecast boundary-layer height and wind. The output states three things together: the two-hour window with the lowest expected exposure, how much lower it is than the day's peak and mean, and **the absolute level in that window with its band for the selected cohort**. If the best window is still in a restrictive band for that cohort, the answer is "No suitable lower-exposure outdoor window identified." The word "safe" is not used.

Measured by regret (exposure in the recommended window minus exposure in the true best window, relative to the daily mean) and by how often the recommendation falls in the true lowest three hours, against the 1.0 seasonal climatology as baseline.

Policy scenarios 2.0 — EP

The simulator stays a **concentration scenario**: "if PM2.5 had been 20% lower", never "if this measure were adopted". Added: up to three scenarios side by side with a baseline; a range for every output, obtained by repeating the scenario by year and season; a weather-normalised baseline before any before/after comparison is described; population weighting where census ward or city populations are available, with the population source cited; and an explicit statement beside the result that pollutants are reduced independently and atmospheric chemistry is not simulated. Linking a named intervention to a concentration change requires a source-apportionment model of the kind the Delhi Decision Support System runs, and is **FW**.

Explanation 2.0 — EP

Attribution is computed for the forecast on screen, not for a stored example. The hundred features are grouped into a few named drivers — recent trend, today's level, weather tomorrow, season and calendar, regional background — and their contributions summed, which also removes the collinearity artefact noted in the audit. The citizen view renders the two largest groups as one sentence generated from a fixed template, so the wording cannot drift from the numbers. The researcher view keeps the full attribution, dependence plots, residuals, calibration and error by band, city and season. Every explanation carries the line: *this describes what the model used, not what causes pollution.*

Trust and failure behaviour — EP / ER

CONDITION	DETECTION	BEHAVIOUR
Station stale or missing	No reading within three hours	Forecast withheld for that station: "No fresh data for this station since 14:00." Nearest valid station or the modelled value offered, labelled
Implausible reading	Range, flat-line and neighbour checks	Reading excluded and flagged; shown in the data-source panel
Input far from training experience	Distance in feature space, assessed **within the predicted band** so that pollution alone does not trigger it	Forecast shown with a caution and the widened range
High pollution with wide range	Predictive distribution	**Never withheld.** Advisory escalates to the cautious quantile
Model drift	Rolling error and rolling interval coverage against their training-time values	Alert to maintainers; coverage figure shown on the trust page
Upstream API failure	Scheduler	Last successful retrieval time shown; nothing fabricated

Part 3 — Platform architecture

Everything in this part is **Proposed**. The single largest change from 1.0 is that the forecast is no longer computed when a user asks for it.

G. AirGuard 2.0 architecture

G.1 Batch first

In 1.0 every page view runs the model inside a serverless function, which is why the function must fit 250 MB, why the classifier had to be swapped for a smaller one, and why the first request takes five seconds. In 2.0 a scheduled job computes tomorrow's forecast for every station once, stores it, and the website reads a row.

```
 Providers            Ingestion (scheduled)         Storage                Batch ML (scheduled)        Serving
 ---------            ---------------------         -------                --------------------        -------
 CPCB / OpenAQ  --->  connector → validate    ---> observations     --->  feature snapshot      --->  read API
 CAMS (Open-Meteo)    → normalise units/time        weather_forecasts      (point-in-time)             (rows
only)
 Weather forecast     → quality flags               cams_forecasts    <--- forecast + interval    --->  Next.js
app
 FIRMS fires          → de-duplicate                feature_snapshots      + cohort advisories
                      → record run                  forecasts              verification (next day)     on-
demand:
                                                    forecast_verification  drift report (weekly)       what-if,
                                                    model_versions         retrain + gate (monthly)


scenarios
```

Consequences: any model that can run in a scheduled job can be used, with no bundle limit; the calibrated classifier problem disappears, because there is no separate serving model; page loads take one database read; and every forecast ever issued is kept, so the system can show how accurate it has been.

Only two things still compute on demand, because they depend on user input: the what-if sliders and the policy scenarios.

G.2 Component choices

COMPONENT	CHOICE	ALTERNATIVES CONSIDERED	WHY	WHAT WOULD CHANGE THE CHOICE
Scheduler	GitHub Actions scheduled workflows	Vercel Cron; database cron; a small always-on server	The pipeline already lives in the repository; free; logs and secrets are built in; independent of the hosting plan	Schedule delays long enough to miss the issue time; inactivity rules on scheduled workflows
Serving database	PostgreSQL (managed free tier)	Static JSON in the repository (1.0); TimescaleDB; SQLite	Concurrent writes, time-range queries, constraints that prevent duplicates. Volume is small: a few hundred stations hourly	Row counts beyond the free tier; then partitioning or TimescaleDB
Training store	Parquet snapshots in object storage, hashed	Training directly from the serving database	Training data must be immutable and reproducible; a live table is neither	—
Read API	Route handlers in the existing Next.js app	Separate FastAPI service	One deployment; reads are trivial	A need for Python on the read path
Compute API	The two existing Python functions	Container service	Already working and verified	Models that cannot be evaluated without ML libraries
Model registry	A `model_versions` table plus hashed artefacts	MLflow	No server to run; enough for one model family	Many concurrent experiments
Maps	Vector tiles with a light open-source renderer	Embedded third-party map	Control over severity encoding and accessibility	—

P. Backend and API

P.1 Endpoints

All under `/api/v1`. Reads are `GET` and cacheable.

ENDPOINT	RETURNS
`GET /cities`	Covered cities with station counts and data freshness
`GET /stations?city=`	Stations with coordinates, source, last reading time, quality status
`GET /live?city=` or `?station=`	Latest valid measurement per pollutant, each with source, `observed_at`, `retrieved_at`, quality flag, `is_modelled`
`GET /forecast?city=&cohort=`	Tomorrow's expected AQI, range, band probabilities, dominant pollutant, cohort advisory, confidence word, issue time, model version
`GET /forecast/history?city=&days=`	Past forecasts beside what happened: the system's own track record
`GET /advisory?city=&cohort=&lang=`	Advisory text with the published source of each sentence
`GET /outdoor-window?city=&cohort=`	Lowest-exposure window, relative reduction, absolute level, or "none identified"
`POST /policy/simulate`	Scenario result with sensitivity range
`POST /forecast/what-if`	Counterfactual forecast for adjusted inputs
`GET /model/explanation?forecast_id=`	Grouped driver contributions and the plain-language sentence
`GET /model/health`	Rolling error, rolling interval coverage, drift status, last retrain, last ingestion run
`GET /data/provenance`	Sources, licences, attribution text, last successful retrieval per source

P.2 Contract rules

- **Every response says how fresh it is**: `data_as_of`, `issued_at`, `model_version`, `sources`.
- **Errors are typed**, in one envelope, with correct status codes: 400 malformed request, 404 unknown city, 422 out-of-range value, 503 data unavailable with `retry_after`. The 1.0 behaviour of returning 500 with an exception string on malformed JSON is removed.
- **Inputs are validated against a schema** at the boundary. What-if adjustments are bounded to the range the interface offers.
- **Caching**: live data five minutes; a forecast until the next issue time.
- **Rate limiting** per client on the compute endpoints; CORS restricted to the application's own origin for `POST`.
- **Versioning** in the path; a breaking change is a new version.

```
{
  "city": "Chennai",
  "forecast_for": "2026-10-08",
  "issued_at": "2026-10-07T12:30:00Z",
  "data_as_of": "2026-10-07T12:00:00Z",
  "aqi": { "expected": 92, "range_80": [74, 118] },
  "band_probability": { "Satisfactory": 0.61, "Moderate": 0.34, "Good": 0.05 },
  "dominant_pollutant": "PM10",
  "confidence": "moderate",
  "advisory": { "cohort": "asthma_copd", "band": "Moderate", "issued_from_quantile": 0.85,
                "target_under_warning_rate": 0.05 },
  "model_version": "2026.10-lgbm-cqr-3",
  "sources": ["OpenAQ (originating from CPCB)", "CAMS via Open-Meteo", "Open-Meteo forecast"]
}
```


The numbers in this example are illustrative of the format only.

Q. Database schema

```
create table sources (
  source_id      text primary key,           -- 'openaq', 'cams_open_meteo', ...
  display_name   text not null,
  licence        text not null,
  attribution    text not null,
  is_official    boolean not null,           -- true only for a government feed
  is_modelled    boolean not null
);

create table stations (
  station_id     bigserial primary key,
  source_id      text references sources,
  source_key     text not null,              -- the provider's own identifier
  name           text, city text, state text,
  latitude       double precision, longitude double precision,
  first_seen     timestamptz, last_seen timestamptz,
  unique (source_id, source_key)
);

create table observations (
  station_id     bigint references stations,
  pollutant      text not null,
  averaging      text not null,              -- 'hour', 'day'
  observed_at    timestamptz not null,
  value          double precision,
  unit           text not null,
  retrieved_at   timestamptz not null,
  quality_flag   text not null,              -- 'ok','stale','flatline','out_of_range','spike'
  primary key (station_id, pollutant, averaging, observed_at)
) partition by range (observed_at);

create table weather_forecasts (
  latitude double precision, longitude double precision,
  variable text, valid_at timestamptz, run_at timestamptz,   -- run_at makes lead time explicit
  value double precision, source_id text references sources,
  primary key (latitude, longitude, variable, valid_at, run_at)
);

create table feature_snapshots (
  snapshot_id    bigserial primary key,
  station_id     bigint references stations,
  issue_time     timestamptz not null,
  feature_version text not null,
  features       jsonb not null,             -- exactly what the model saw; never rewritten
  unique (station_id, issue_time, feature_version)
);

create table model_versions (
  model_version  text primary key,
  trained_from   date, trained_to date,
  data_hash      text not null, feature_version text not null,
  hyperparameters jsonb, metrics jsonb,
  artefact_uri   text not null, artefact_sha256 text not null,
  deployed_at    timestamptz, is_active boolean not null default false
);

create table forecasts (
  forecast_id    bigserial primary key,
  station_id     bigint references stations,
  issue_time     timestamptz not null,
  target_date    date not null,
  model_version  text references model_versions,
  snapshot_id    bigint references feature_snapshots,
  expected       double precision, quantiles jsonb, band_probability jsonb,
  status         text not null,              -- 'issued','withheld_stale_data','low_confidence'
```

```
  unique (station_id, target_date, model_version)
);

create table forecast_verification (
  forecast_id    bigint primary key references forecasts,
  observed       double precision, absolute_error double precision,
  inside_interval boolean, band_error integer, verified_at timestamptz
);

create table ingestion_runs (
  run_id bigserial primary key, source_id text references sources,
  started_at timestamptz, finished_at timestamptz,
  status text, rows_written integer, error text
);
```

`alert_subscriptions`, `alert_events` and `evaluation_results` are added with the alerting and research-lab features.

R. MLOps

CADENCE	JOB	OUTPUT	GATE
Hourly	Ingest, validate, flag	`observations`, `ingestion_runs`	Retries with exponential back-off; failure recorded, never papered over
Daily, issue time	Feature snapshot → forecast → advisories	`feature_snapshots`, `forecasts`	Stations with stale data are marked withheld
Daily, next morning	Verify yesterday's forecast	`forecast_verification`	—
Weekly	Drift report: rolling MAE, rolling coverage, feature distribution shift	`/model/health`	Alert if coverage falls more than five points below nominal
Monthly	Retrain a challenger on the extended window; rolling-origin evaluation	Candidate in `model_versions`	**Promotion needs a person to approve** it, and requires the challenger to be no worse on *Poor*-or-worse days
On demand	Rollback	`is_active` flipped to the previous version	One command

Recorded for every model: version, training window, data hash, feature version, hyperparameters, metrics, artefact hash, deployment time.

Reproducibility

Retained from 1.0: one command rebuilds everything; seeds are fixed; no reported number is typed by hand. Added: a locked environment file; configuration separated from code; a data snapshot hash in every result file; and a `paper` build target that regenerates every table and figure in the manuscript from result files, so the paper cannot disagree with the experiments.

Security and privacy

- **No account and no health record is needed** to use the product. The cohort is chosen from seven broad options and kept on the device.
- If alerts are added, the only personal datum is a contact address, stored with consent, encrypted at rest, and deletable by the user.
- Secrets live in the scheduler's and host's secret stores, never in the repository. Provider sample keys are not used in production.
- Response headers: Content-Security-Policy, X-Frame-Options, X-Content-Type-Options, Referrer-Policy (1.0 sends only HSTS).
- Dependencies are audited in continuous integration; 1.0 already moved off a Next.js release with a published vulnerability.
- The advisory is separated from clinical advice in the interface and in the API, and every advisory carries the disclaimer.

Part 4 — Product and interface

Everything in this part is **Proposed**. Per the implementation order, none of it is built before the data and model work in Parts 2 and 3; an interface designed first would be designed around numbers that do not exist yet.

M. Interface redesign specification

M.1 Honest assessment of the current interface

The 1.0 interface was restyled for visual impact: a dark background with coloured radial gradients, translucent blurred panels, gradient-filled headline text, a large hero block, and symbol characters used as icons. That is the aesthetic the brief now rules out, and the brief is right to. It signals "demo", it spends contrast on decoration, and it gives the same visual weight to a headline metric and a caveat. It is replaced, not refined.

Two things from 1.0 are kept: tabular numerals for every figure, and the colour-blind-safe palette for data.

M.2 Design personality

An instrument, not a poster. The reference points are a good weather product (now, then next, then detail), a financial terminal (density without clutter, numbers aligned to be compared), a public-health dashboard (plain words, caveats beside the figure), and a government monitoring portal (provenance always visible). These are patterns to learn from; no layout, branding, illustration or code is copied from any product.

PRINCIPLE	IN PRACTICE
The number is the hero	Tomorrow's AQI and its range are the largest elements on any screen that shows them
Uncertainty is drawn, not footnoted	Every forecast appears with its range; a forecast without a range is not rendered
Provenance is always in view	Source and "as of" time sit beside every measurement
One accent, used for interaction only	Colour otherwise belongs to data
Calm	Flat surfaces, hairline rules, no glow, no gradient fills, no blur, no decorative animation
Progressive disclosure	A resident sees four things; a researcher can reach everything

M.3 Personas, and the navigation tested against them

PERSONA	NEEDS IN THE FIRST TEN SECONDS	RARELY NEEDS
Resident or parent — child with asthma, Chennai	Is tomorrow a problem for my child, and when is the least bad time to go out?	Model diagnostics
Planner — school sports coordinator, site supervisor	Should tomorrow's outdoor activity move, and to when?	Per-pollutant detail
Policy analyst — city or state official	How does this city compare, and what would a reduction change?	Individual advice
Researcher, student, judge	How do you know, how accurate, on what data, and what does it fail at?	Simplified wording

The brief suggests eleven navigation labels. Tested against these personas, several collapse: an "Exposure Advisor" is not a place a resident goes, it is a property of the forecast they are already reading; "Alerts" is an action; "Model Trust", "Research Lab" and "Data Sources" serve one persona and belong together. Proposed structure:

SECTION	REPLACES FROM THE BRIEF	PRIMARY PERSONA
Today	Command Center, Live Air, Exposure Advisor	Resident, planner
Forecast	Tomorrow Forecast	Resident, planner
Map	India Risk Map, City Intelligence	Analyst, resident
Policy Lab	Policy Lab	Analyst
Evidence	Research Lab, Model Trust, Data Sources	Researcher

Alerts are a control in the header. The cohort and city selectors are global and persist. This structure is a hypothesis: it is to be checked with a first-click test on five people per persona before it is fixed.

N. Page wireframes

Described top to bottom. Desktop uses a 12-column grid; the mobile order is given where it differs.

N.1 Today

Answers: what is happening now, what is likely tomorrow, who should be careful, why, and what to consider doing.

REGION	CONTENT	NOTES
Header bar	Product name · city selector · cohort selector · language · alerts	Selectors are global
Status line	"Measured at Alandur, Chennai · 8 minutes ago · source: OpenAQ (CPCB)"	Turns to a warning state when stale: "No fresh data since 14:00"
Now (4 columns)	Current AQI, category label and level, dominant pollutant, 24-hour sparkline	Measured and modelled values are visually distinct
Tomorrow (5 columns)	Expected AQI at display size · range bar · band probability bar · confidence word · trend against today	The range bar is part of the number, not a tooltip
For you (3 columns)	Advisory for the selected cohort: band, one action sentence, one cohort sentence	Each sentence links to its published source
Lowest-exposure window	24-hour strip for tomorrow with the recommended two hours marked, reduction against the peak, and the absolute level in that window	Reads "No suitable lower-exposure window identified" when that is the truth
Why	One sentence naming the two largest drivers; link to full explanation	—
Track record	"Last 30 days: typical miss 14 AQI points; range contained the outcome on 27 of 30 days"	Read from stored verification
Footer	Sources, licences, disclaimer	Always present

Mobile order: status line, Tomorrow, For you, Window, Now, Why, Track record. Tomorrow comes before Now because it is the decision.

N.2 Forecast

REGION	CONTENT
Selectors	City or station · cohort · date (tomorrow by default; past dates open the stored forecast beside the outcome)
Headline	Expected AQI, range, band probabilities, confidence, model version, issue time
Trajectory	Fourteen days of observed AQI, then tomorrow's range drawn as a band, with category guide lines labelled on the axis
Hourly	Tomorrow hour by hour with uncertainty band; window marked
Pollutants	Per-pollutant forecast with sub-index; dominant pollutant flagged
Weather tomorrow	Wind, boundary-layer height, rain, with a one-word reading of each ("stagnant", "ventilated")

REGION	CONTENT
All cohorts	Table: cohort, advisory band, the quantile it was issued from, the target under-warning rate
Why this forecast	Grouped drivers as a horizontal bar list, plain sentence above; expandable to full attribution
What-if	Sliders, clearly marked counterfactual, bounded
Days two and three	Shown only where validated; otherwise absent, not greyed

N.3 Map

REGION	CONTENT
Map	India, station markers. Toggle: now · tomorrow expected · tomorrow upper bound · dominant pollutant · cohort advisory
Marker	Shape and number carry severity, colour reinforces it. Stale stations are hollow
Side panel	Ranked list of cities for the active layer; sortable; search
Drill-down	City → stations → station page with history and forecast
Layer honesty	Tomorrow layers appear only for stations with a valid forecast; others show "no forecast"

N.4 City

Seasonal pattern by month, station list with freshness, track record for that city, events (festival windows, described without causal language), archetype if the station-level clustering supports one.

N.5 Policy Lab

REGION	CONTENT
Framing	"A concentration scenario. It does not predict the effect of any measure." Fixed, not dismissible
Scenario builder	Up to three scenarios beside a baseline; per-pollutant reduction; scope by city, season, year
Results	For each scenario: AQI change, days moving band, cohort advisory-days, each **with a range**
Comparison	Small-multiple bars, common axis; table beneath
Method	What is and is not simulated; the weather-normalisation note

N.6 Evidence

Tabs: **Track record** (rolling error, rolling coverage, error by band, city and season — live from verification) · **Model** (version, training window, features, benchmark table, ablation) · **Calibration** (reliability diagram, interval coverage, under- and over-warning by cohort) · **Generalisation** (leave-one-city-out, excluded cities) · **Data** (sources, licences, missingness, quality flags, last retrieval) · **Method** (data card, model card, advisory method) · **Limitations** (the list in Part 5, in full).

N.7 States every page must design for

STATE	TREATMENT
Loading	Skeleton in the final layout
No fresh data	Named station, time of last reading, modelled value offered and labelled
Forecast withheld	Reason given in a sentence; never an empty panel
Low confidence	Forecast shown, range emphasised, advisory from the cautious quantile
Upstream outage	"Data temporarily unavailable", last successful retrieval time
Historical replay	A persistent band across the page: "Historical data, 2015–2020 — not a live forecast"

O. Design system

O.1 Foundations

TOKEN	SPECIFICATION
Typeface	One humanist or neo-grotesque sans for interface and prose; tabular lining numerals everywhere; a monospace only in the Evidence section
Type scale	12 · 14 · 16 (body) · 20 · 28 · 40 · 72 (the AQI numeral)
Line length	60–75 characters for prose
Spacing	4-point base: 4 · 8 · 12 · 16 · 24 · 32 · 48 · 64
Grid	12 columns to 1280 px; 8 columns tablet; 4 columns mobile; 24 px gutters desktop, 16 px mobile
Surfaces	Light theme first: paper-neutral background, white panels, one-pixel hairlines. A true dark theme is a second token set, not an inversion
Elevation	None beyond a hairline; a shadow only on floating elements

TOKEN	SPECIFICATION
Radius	4 px controls, 8 px panels
Accent	A single hue reserved for links, focus and selected controls
Motion	State changes only, 150 ms or less; nothing moves under reduced-motion

O.2 Severity system

Severity is the most important encoding in the product and must survive greyscale printing, colour-vision deficiency and a screen reader.

CHANNEL	RULE
Label	The CPCB category name is always present in text
Level	A number from 1 to 6 and a matching six-segment glyph accompany the label
Colour	An ordered ramp that remains monotone in lightness, checked for the common colour-vision deficiencies; the official CPCB category colour is shown as a small reference chip so the product agrees with official communication
Shape	Map markers change shape as well as colour above "Poor"
Text contrast	4.5 : 1 minimum on every severity background

O.3 Components

Stat block · range bar · band-probability bar · freshness badge · source tag · cohort switch · advisory card · driver list · 24-hour window strip · track-record line · station marker · data table with sortable, sticky headers · scenario comparison · status banner · unavailable state. Each is specified with its empty, loading, stale and error variants before it is built.

O.4 Chart rules

One message per chart, stated in its title as a sentence. Direct labels in place of legends where possible. Uncertainty always drawn. Bars start at zero. Category guide lines labelled on the axis. No dual axes, no three-dimensional effects, no decorative fills. Every chart has a table equivalent reachable by keyboard.

O.5 Iconography

One line-icon set at a single stroke weight. No symbol characters used as icons. An icon never carries meaning alone.

O.6 Accessibility

Keyboard reachability of every control with a visible focus ring; landmarks and labelled regions; severity announced as "AQI 187, Moderate, level 3 of 6"; live regions for data refresh; text resizable to 200% without loss; touch targets 44 px; a high-contrast token set. Target: WCAG 2.2 AA, checked with automated tools and one manual screen-reader pass per release.

O.7 Languages

All strings externalised from the first commit of 2.0; layout tolerant of longer strings and of Tamil and Devanagari line heights. English, Tamil and Hindi first. The 1.0 Streamlit application carries advisory text in all three, but the Tamil and Hindi strings were machine-drafted and have not been reviewed by a fluent health communicator, and the web application dropped them. **Health advice is not machine-translated**: each translation is reviewed by a fluent speaker with health-communication experience before release, and an unreviewed language is not offered.

U. Startup and SaaS roadmap

Monetisation follows proof, not the other way round.

STAGE	GOAL	OFFER	EVIDENCE THAT IT IS WORKING
0 — Credibility	Show the forecast is accurate, in public	Free site with a live track-record page	Thirty days of verified forecasts, published
1 — Citizen product	Useful daily for residents	Today, Forecast, Map, alerts, three languages	Return visits; alert opt-ins
2 — Pilots	One real organisational user in each of two segments	Schools: outdoor-activity planner for the week. Employers of outdoor workers: shift-exposure view	A signed pilot; a decision actually changed because of the product
3 — Services	Sustainable revenue	Forecast and alert API; organisation dashboards; research workspace	Paying customers

Segments worth testing, in order of fit to what the system does well: schools; employers with outdoor workforces; hospitals and public-health communicators; municipal and smart-city teams; universities and NGOs.

Constraints that must be resolved before any revenue: the weather source's free tier is non-commercial; each station-data source's terms must be read for redistribution; advisory content needs professional review and clear limits of liability; and none of the vulnerable-group claims may be marketed as medical.

V. Competition demonstration flow

Each scene is marked with what exists today. A demonstration given before the build must use only the scenes marked as existing, and say so.

#	SCENE	SHOWS	STATUS TODAY
1	Select Chennai: live AQI, station, minutes since reading	Measurement with provenance	**Proposed** — 1.0 has no live data
2	Tomorrow's AQI with its range and band probabilities	Uncertainty as a first-class output	**Proposed**
3	Switch healthy adult → child → asthma	Advice differs, and each is issued at a stated risk level	**Partial** — cohort switch exists; risk level does not
4	Tomorrow's lowest-exposure window, with the absolute level	Honest relative guidance	**Partial** — seasonal climatology only
5	"Why this forecast?"	Plain sentence, then the driver list	**Partial** — fixed example only
6	Map of tomorrow's risk; toggle to the upper bound	Geography of uncertainty	**Proposed**
7	Policy Lab: PM2.5 lower by 20%; advisory-days by cohort, with ranges	Scenario, not prediction	**Implemented** without ranges
8	Evidence: thirty-day track record, coverage, drift, limitations	The system reporting on itself	**Proposed**

The line to deliver over scene 8: *measurement, prediction, uncertainty, a decision made at a stated risk, and a public record of how often it was right.*

A demonstration that is honest today uses scenes 3, 5 and 7 on the historical replay, then shows the pilot tables P1–P5 as the motivation for 2.0: what the current system gets wrong, measured, and the experiment designed to fix it. For a research audience that is a stronger story than a polished interface.

Part 5 — Programme

S. Implementation roadmap

Estimates assume a team of three working part-time. The order follows the brief: evidence, then data, then models, then the product. Interface work does not start until the numbers it will display exist.

PHASE	WEEKS	WORK	EXIT CRITERION	CLASS
0 — Evidence	done	Audit, prior-art corpus, five pilot experiments, this blueprint	This document	ER
1 — Corrections	1	Fix the seven items in Part 1 A.3 on the live site and in the reports	No statement on the site exceeds its evidence	EP
2 — Data access	1	Register keys; confirm the official feed; decide the measured-data source	A real station reading retrieved and stored with full provenance	ER
3 — Ingestion	1–3	Connectors, validation, quality flags, storage, scheduler	Seven days unattended, ≥ 95% of runs successful, staleness visible	EP
4 — Backfill	2–4	Station history from 2020; archived forecasts from 2024; CAMS from 2022; feature snapshots	A training table in which every feature is provably available at issue time	ER
5 — Benchmark	4–6	Reference forecasts, model tiers, ablation	H2 and H4 answered with confidence intervals	ER
6 — Uncertainty and advisory	6–8	Predictive distribution, conditional coverage, risk-controlled advisory, unseen-city tests	H1, H3 and H5 answered	ER
7 — Advisory index, window, scenarios	7–9	Index sensitivity and anchoring; hourly window; scenario ranges	Sensitivity table published; window beats climatology or is withdrawn	ER / EP
8 — Serving	8–10	Batch forecasts, read API, daily verification, drift report	Forecast for every station by issue time; yesterday verified each morning	EP
9 — Interface	9–12	New information architecture, design system, accessibility, three languages	First-click test passed; WCAG 2.2 AA audit passed	EP

PHASE	WEEKS	WORK	EXIT CRITERION	CLASS
10 — Track record and paper	12–14	Thirty days of live verified forecasts; dedicated novelty search; manuscript	Paper tables regenerate from result files	ER
Later	—	Schools and workforce pilots; API; satellite and traffic inputs; health-outcome validation	—	HV / FW

One thing is time-sensitive. The pilot's weakest point is the sample of severe days. Northern India's post-monsoon and winter pollution season begins within weeks. Ingestion started now captures it; ingestion started in January does not. Phases 2 and 3 should begin immediately, ahead of everything else.

T. Research paper outline

The manuscript is written only after Phases 5–7 produce results. No figure in it is typed by hand.

1. Abstract
2. Introduction — the two-sided error of an air-quality advisory
3. Problem statement — advisories as decisions under uncertainty
4. Related work — operational forecasting in India; health-based indices; impact-based warning; uncertainty and conformal methods; cost-loss decision models
5. Research gap — from the reviewed corpus, with its scope stated
6. Contributions
7. Data sources and provenance
8. Data quality and preprocessing
9. Feature engineering and the prediction-time contract
10. Forecasting architecture and benchmark
11. Vulnerability-aware advisory method — two layers, risk-controlled issuance
12. Uncertainty estimation and conditional coverage
13. Scenario analysis
14. Explanation
15. Experimental design — rolling origin, unseen cities, regime slices
16. Baselines — persistence, climatology, CAMS, one-band shift
17. Results
18. Ablation
19. Cross-city generalisation
20. Error analysis, with the abstention finding

21. Discussion
22. Limitations
23. Ethical considerations — over-warning fatigue, unreviewed health text, who sets the risk level
24. Future work
25. Conclusion

Which of the brief's candidate contributions survive

CANDIDATE	VERDICT ON THE EVIDENCE	FORM IT MAY TAKE
C1 Real-time multimodal next-day forecasting for India	Not novel: operational systems exist	A measured comparison against CAMS, nothing more
C2 Uncertainty-aware prediction	Not novel alone	**Conditional coverage on extreme days** as the evaluated quantity
C3 Vulnerability-aware advisory integrated with forecasts	Exists (Forecaster.health, tiered official advice)	Only through risk-controlled issuance
C4 Dynamic lower-exposure window	Exists commercially	Product feature
C5 Unified forecast → explanation → scenario workflow	Integration	Product contribution
C6 Cross-city generalisation and trust monitoring	Protocol exists; pilot effect is small	Reported honestly; the abstention analysis is the interesting part

Retained for the paper: one methodological contribution (risk-controlled cohort advisories, H1) and four experimental ones (conditional coverage on extremes; evidence that uncertainty-based abstention silences severe days; the value of forecast meteorology at true lead time; skill against CAMS and in unseen cities), with an open, reproducible pipeline and a public track record as the artefact.

Candidate venue types, to be matched to the strength of the results rather than chosen now: environmental modelling and atmospheric-science journals for the forecasting and evaluation; applied machine-learning and AI-for-social-impact tracks for the decision method; a climate-and-AI workshop as an early, low-risk route for the pilot findings.

W. Risks and mitigation

RISK	LIKELIHOOD	EFFECT	MITIGATION
No reliable, permitted source of live station measurements	Medium — the official API could not be verified	Product cannot claim measured live data	Confirm in Phase 2 before anything else; fall back to modelled CAMS, labelled as modelled; never relabel
Too few severe days for firm conclusions	High	Weak claims on the cases that matter	Start ingestion before winter; pool stations; report intervals; do not over-read
Weather adds less than expected	Medium	H2 negative	Report it; the advisory method does not depend on it
Risk-controlled rule does not beat the one-band shift	Medium	H1 negative	Pre-declared; the simpler rule is adopted and the comparison is the result
Guarantee fails under drift	Medium	Under-warning exceeds target in some seasons	Rolling calibration; monitor realised rate; show it publicly
Advisory index challenged as arbitrary	High	Credibility	Two-layer design; sensitivity analysis; professional review; the word "validated" is never used
Over-warning fatigue	Medium	People stop listening	Over-warning is measured and shown beside under-warning
Health text causes harm or liability	Low, severe	Serious	Published-source tiers only; reviewed translations; disclaimer; no diagnosis
Free-tier and licence limits	Medium	Outage or breach of terms	Terms read before use; attribution automated; commercial tier before revenue
Scope exceeds a student team	High	Nothing finished	The roadmap is ordered so that stopping after Phase 6 still yields a complete paper
Concept drift	Certain	Accuracy decays	Daily verification, weekly drift report, monthly retrain with a human gate

Limitations, stated in advance

- Live coverage will be uneven: stations fail, report late, and vary in quality.
- A station represents its location, not the exposure of everyone in a city.
- Weather forecasts are themselves uncertain, and that uncertainty passes into the air-quality forecast.
- Extreme episodes remain the hardest cases and have the least data.
- The advisory index is evidence-informed and not validated against health outcomes.
- Scenario results are arithmetic on concentrations; they predict no intervention.
- The CAMS grid is about 45 km and cannot see a neighbourhood.
- Satellite columns are not surface concentrations and are missing under cloud.
- Lockdown and festival comparisons are descriptive until weather-normalised.
- The prior-art review covers 47 sources and supports "among the systems reviewed", not "first".

X. Acceptance criteria

The sixteen questions the brief requires the system to answer, each with the test that decides whether it does. A question the system cannot answer responsibly is shown as future work rather than answered badly.

QUESTION	ACCEPTED WHEN
What is the air quality right now?	A measured value with station and source, or an explicit "unavailable"
How fresh is that measurement?	Age shown beside every value; stale after three hours
What is tomorrow's predicted AQI?	Issued daily for every covered station by the issue time
What is the uncertainty?	A range whose realised coverage is within five points of nominal over the last 30 days
Which category is most likely?	Band probabilities with calibration error ≤ 0.03
Which pollutant is driving it?	Dominant pollutant computed from the forecast concentrations
Why did the model predict this?	Grouped drivers for that forecast, in a sentence generated from them
How does advice differ by group?	Advisory per cohort with its risk level; realised under-warning at or below target
When might exposure be lower?	A window that beats the climatological one on held-out days, with the absolute level, or "none identified"
How does this city compare?	Ranked map layer from the same forecasts

QUESTION	ACCEPTED WHEN
What happens under a reduction?	Scenario with a range and the non-causal statement
How accurate has it been on unseen future data?	A public rolling track record from stored forecasts
Does it generalise?	Unseen-city and excluded-city results published, including the failures
Is the model drifting?	Rolling error and coverage against reference, with a stated alert threshold
What source produced this?	Source, licence and retrieval time on every value
What are the limitations?	The list above, in the product, one click from every page

Decisions needed before Phase 2

1. **API keys.** Register on data.gov.in, OpenAQ and NASA FIRMS (each free), and confirm whether the official CPCB feed is actually available. This decides whether the product may say "measured".
2. **Database.** Approve a managed PostgreSQL free tier, and authorise the connector for it.
3. **Corrective deploy.** Approve pushing the seven corrections in Part 1 A.3 to the live site.
4. **Target and date.** Which competition or venue comes first, and when. It sets where the roadmap is cut.
5. **A reviewer.** One clinician or public-health professional willing to read the advisory text, and one fluent reviewer each for Tamil and Hindi.